RUSHIKESH MOHALKAR

AI/ML ENGINEER · DEEP LEARNING & AGENTIC AI

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SUMMARY

AI/ML Engineer with a core interest in **deep learning**, from training Transformers from scratch in PyTorch to building production **Agentic AI** systems (**multi-agent workflows**, tool calling, RAG, guardrails, AI evaluation). Strong Python and backend skills (FastAPI, Docker) with LangChain and LangGraph, backed by 2+ years of hands-on experience.

TECHNICAL SKILLS

Languages: Python, Java, SQL, Bash
Deep Learning: PyTorch, TensorFlow, Keras, Transformers, Self-Attention, CNNs, RNNs, LSTM, Transfer Learning, QLoRA, Mixed Precision (AMP)
GenAI & Agents: LLMs, RAG, LangChain, LangGraph, LlamaIndex, LiteLLM, Hugging Face, Ollama, Prompt Engineering
ML & NLP: Scikit-learn, Feature Engineering, Model Evaluation, NER, Text Classification, Sequence Modeling
Retrieval & Data: FAISS, Qdrant, Embeddings, Reranking, BM25, PostgreSQL, SQLite, Redis
MLOps & Cloud: FastAPI, Flask, Docker, Nginx, Jenkins, CI/CD, Git, Linux, AWS (EC2, S3), MLflow, Prometheus

PROJECTS

Enterprise RAG & AI Operations Platform | *LangGraph, FastAPI, Qdrant, PostgreSQL, Redis* GitHub

- Agentic RAG with hybrid dense + BM25 retrieval, reranking, HyDE, Corrective RAG and Self-RAG (0.76 Precision@5); Text-to-SQL with human-approval via LangGraph interrupts.
- Nine-layer input/output guardrails, semantic caching, and AI evaluation (retrieval metrics, guardrail accuracy) backed by a 160-test suite.

Custodian: Multi-Agent Accounts Payable AI | *LiteLLM, FastAPI, React, MLflow, Prometheus* GitHub

- Multi-agent system with tool calling for invoice ingestion, fraud/risk scoring, approval routing and payments, across OpenAI, Groq, Anthropic and Hugging Face via LiteLLM.
- AI security controls: PII redaction, policy enforcement, audit logging, role-based API auth; 95 backend and 33 frontend tests.

rag-engine-v2: Offline PDF Question Answering | *LangChain, FAISS, Transformers, RapidOCR* GitHub

- Built a production-ready RAG application for PDF Q&A using Python, LangChain, FAISS, Sentence Transformers, and LLMs, enabling efficient semantic retrieval and answer generation from enterprise documents.
- Developed end-to-end document intelligence workflows with Flask UI, PDF/OCR processing, Docker deployment, and AWS Bedrock integration for scalable local and cloud inference.

Oryza: Self-Hosted LLM Chatbot | *Ollama, Qwen 2.5, Flask, Docker, Nginx* GitHub

- Private offline chatbot serving Qwen 2.5 (1.5B) via Ollama, containerized with Docker Compose and Nginx for resource-constrained deployment.
- Flask UI forwards prompts to the Ollama API for local Qwen inference; Flask, Ollama and Nginx run as separate containers, configurable via environment variables.

NanoChat-X: GPT-Style Transformer from Scratch | *PyTorch, FastAPI* GitHub

- Built a decoder-only Transformer in PyTorch: causal self-attention, token/positional embeddings, weight-tied LM head, and custom tokenizer.
- Training stack with AdamW, warmup-cosine LR, gradient accumulation and AMP; temperature/top-k sampling served via FastAPI, causal masking verified by regression tests.

EXPERIENCE

Cognizant Technology Solutions | *Programmer Analyst (Intern > PAT > PA)* **Jan 2024 – Present**

- Automated the data validation pipeline between a mainframe application and its web counterpart using Python, producing clean reports and cutting manual effort by 50%.
- Built and maintained data-driven workflows and reporting solutions, improving process efficiency and reducing delivery delays by 20%.
- Worked across Python, Java, SQL, Git, Docker, Jenkins, Maven and CI/CD in an Agile enterprise environment.

EDUCATION

B.E. Electronics & Telecommunications Engineering, AISSMS IOIT, Pune | CGPA: 8.21/10 **2019 – 2023**

CERTIFICATIONS

Mathematical Foundations of ML | Python for Data Science & ML | Deep Learning A-Z

AI Engineer Core Track: LLMs, RAG, QLoRA & Agents | GenAI with LangChain & Hugging Face